

Daily Sequences Drill #3

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Geometric Sequences & Series · Sum to Infinity · The Convergence Condition $|r| < 1$.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Ratios, terms, and the sum-to-infinity formula on sight.

- Write down the common ratio of the GP 3, 6, 12, 24, ...
- A GP has first term 5 and common ratio 2. Find the 6th term.
- A GP has first term 2 and common ratio 3. Find S_5 .
- State the condition on r for an infinite GP's sum to converge.
- A GP has first term 8 and common ratio $\frac{1}{2}$. Find its sum to infinity.
- Which of these three sequences is geometric: (i) 2, 4, 6, 8 (ii) 2, 4, 8, 16 (iii) 2, 4, 7, 11?
- Write down the sum-to-infinity formula for a GP with first term a and ratio r (where $|r| < 1$).
- True or false: a GP with common ratio $r = 1$ has a well-defined sum to infinity.
- A GP has first term 100 and common ratio $-\frac{1}{2}$. Find its sum to infinity.
- Write the next term of the GP 81, 27, 9, 3, ...

Section B Manipulation Drills

(≤50 s each)

Solve for the GP's parameters, then use them — always check convergence first.

- A GP has first term 4 and common ratio 3. Find the smallest n for which $S_n > 1000$.
- For which values of r does the infinite GP with first term 5 have a finite sum?
 - all real r
 - $r > 0$ only
 - $-1 < r < 1$

- D) $r \neq 1$
3. A GP has first term 3 and common ratio $\frac{2}{3}$. Find its sum to infinity.
4. A GP has sum to infinity 20 and first term 5. Find r .
- A) $\frac{3}{4}$
B) $\frac{1}{4}$
C) $\frac{4}{5}$
D) $-\frac{3}{4}$
5. A GP has first term 1 and sum to infinity 2. Find the second term.
- A) $\frac{1}{2}$
B) $\frac{1}{4}$
C) 2
D) 1
6. An AP and a GP both have first term 4. The AP has common difference 2; the GP has common ratio 2. Which sequence has the larger 10th term?
- A) The AP
B) The GP
C) They are equal
D) Cannot be determined
7. A GP has all terms positive and common ratio $r > 1$. Does its sum to infinity exist?
- A) Yes, always.
B) No, never — it diverges.
C) Only if the first term is small.
D) Only if $r < 2$.
8. Express the recurring decimal $0.\overline{3} = 0.333\dots$ as a fraction, using the geometric series sum-to-infinity formula.
9. A GP has first term 2 and common ratio -3 . Find S_4 .
- A) 40
B) -40
C) 160
D) -160

10. Recall from Day 2 that $(1 - x)^{-1} = 1 + x + x^2 + \cdots$ for $|x| < 1$. Using $x = \frac{1}{3}$, what does this series sum to?

- A) $\frac{3}{2}$
- B) $\frac{2}{3}$
- C) 3
- D) 2

Section C Substitution & Structure

(≤90 s each)

Convergence first, computation second.

1. A GP has first term 1 and unknown common ratio r . Its sum to infinity exists and equals 4. What must be true about r ?

- A) $r = \frac{3}{4}$ exactly.
- B) $-1 < r < 1$, but no exact value is determinable.
- C) $r = \frac{4}{3}$.
- D) Infinitely many values of r satisfy this.

2. A geometric sequence has first term a and common ratio r , both positive integers with $r > 1$. Let S_n denote the sum of the first n terms. Given $S_{18} - S_{12} = kS_6$ for some positive integer k , find the smallest possible value of k . (after TMUA 2022 Paper 1 Q8)

- A) 2^6
- B) 2^{12}
- C) 2^{18}
- D) $\frac{2^6}{2^6 - 1}$
- E) $2^6(2^6 - 1)$

3. An infinite geometric series has first term $a = 6$ and common ratio $r = \frac{3k}{10}$, where k is chosen uniformly at random from the integers $-5, -4, \dots, 4, 5$ (11 equally likely values). Find the probability that the series converges *and* its sum to infinity exceeds 5. (after TMUA 2023 Paper 1 Q18)

- A) $\frac{4}{11}$
- B) $\frac{5}{11}$
- C) $\frac{7}{11}$
- D) $\frac{3}{11}$

4. The series $\sum_{k=1}^{\infty} ar^{k-1}$ converges (for real a, r) when $|r| < 1$. If a and r are instead allowed to be complex numbers, which is the correct convergence condition?

- A) $|r| < 1$ (the same condition, using the complex magnitude).
- B) $r < 1$ (a real inequality).

- C) $a < 1$.
- D) There is no meaningful convergence condition for complex r .
5. Convert the recurring base-2 number $0.\overline{0111} = 0.0111\,0111\,0111\ldots$ (base 2) to a fraction in base 10. (after TMUA 2023 Paper 2 Q15)
- A) $\frac{1}{3}$
- B) $\frac{1}{5}$
- C) $\frac{7}{15}$
- D) $\frac{7}{16}$
6. Which of the following statements about arithmetic and geometric series is FALSE?
- A) An AP's terms grow linearly in n ; a GP's terms with $|r| > 1$ grow exponentially in n .
- B) An AP diverges to $\pm\infty$ (or stays constant) as $n \rightarrow \infty$, unless $d = 0$.
- C) A GP can converge to a finite sum to infinity even though no individual term is exactly zero, provided $|r| < 1$.
- D) Every convergent GP has all its terms equal to zero eventually.
7. Using $(1-x)^{-1} = \sum_{k=0}^{\infty} x^k$ from Day 2, evaluate $\sum_{k=0}^{\infty} \frac{1}{5^k}$ by choosing an appropriate value of x .
- A) $\frac{5}{4}$
- B) $\frac{4}{5}$
- C) 5
- D) $\frac{1}{4}$
8. Let $S = 1 + 2x + 3x^2 + 4x^3 + \cdots$. Recall from Day 2 that this is exactly $(1-x)^{-2} = \sum_{k=0}^{\infty} (k+1)x^k$. Evaluate S at $x = \frac{1}{4}$.
- A) $\frac{16}{9}$
- B) $\frac{4}{3}$
- C) $\frac{9}{16}$
- D) $\frac{3}{4}$

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant route — find it.

1. Prove that $\sum_{k=1}^{\infty} \frac{1}{2^k} = 1$ exactly (not merely “approaches 1”), using the sum-to-infinity formula.
- Then explain why the partial sum $S_n = \sum_{k=1}^n \frac{1}{2^k} = 1 - \frac{1}{2^n}$ never actually equals 1 for any finite n .

2. Let $x = 0.\overline{9} = 0.999\dots$ (a recurring 9). Writing x as an infinite geometric series and summing it, determine x exactly.
- A) $x = 1$ exactly.
 - B) x is infinitesimally less than 1 but not equal to it.
 - C) $0.\overline{9}$ is undefined.
 - D) $x = \frac{9}{10}$.
3. A ball is dropped from height h . Each time it bounces, it rebounds to a fraction r of its previous height ($0 < r < 1$). Find, in terms of h and r , the total distance the ball travels (up and down, forever) before coming to rest.
- A) $h \cdot \frac{1+r}{1-r}$
 - B) $h \cdot \frac{2}{1-r}$
 - C) $h \cdot \frac{1}{1-r}$
 - D) $h(1+2r)$
4. The sum to infinity of a GP equals the sum of its first two terms. What can be said about the common ratio r ?
- A) $r = 0$ (the “GP” is actually constant).
 - B) $r = \pm 1$.
 - C) $r = \frac{1}{2}$.
 - D) No such r exists.
5. A sequence is defined by $a_1 = 3$ and $a_{n+1} = \frac{1}{2}a_n$ for $n \geq 1$. Using the sum-to-infinity formula, evaluate $\sum_{n=1}^{\infty} a_n$.
- A) 6
 - B) 3
 - C) 4.5
 - D) It cannot be summed, since it’s defined by a recurrence, not a formula.

“Infinity is not a number and $r = 1$ is not a ratio you may divide by. Two mistakes, one line.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths